Topic 2: Algebra

2.1 Expressions and formulae

Subject content	Guidance	Curriculum reference
A Write expressions using correct algebraic notation		A7.1A A7.1B
B Simplify algebraic expressions by collecting like terms	$6a + 7 + 3a^2 - 4a - 3 + 5a^2$	A7.1C
C Simplify algebraic expressions involving multiplication and division	Write $\frac{4pq \times 5p \times 6q}{12q}$ in its simplest form	A7.1D
D Expand brackets and simplify expressions involving a bracket and/or powers	$5 - 2(d - 5)$ $x(x^2 + 3x + 5)$	A7.1E A8.1C A9.1E
E Factorise expressions	$2x - 8$ $8st - t^2$ $4a^2 + 12a$ $24xy^2 - 6y^4$	A8.1D A9.1E
F Substitute values into expressions and formulae and/or including powers, roots and brackets	If $a = 3$ and $b = 4$, evaluate: $a^2 + \sqrt{9b} - (ab + 3)$	A7.1F A7.1G A8.1E A9.1A
G Solve problems involving formulae and expressions	A power company uses the formula: $C = 0.25h + 0.15r + 30$ to calculate the total cost, C, of their customers' bills where h = number of units used at the higher rate r = number of units used at the regular rate If a customer uses 175 units at higher rate and 250 units at the regular rate, what will be the total costs of their bill?	A7.1I A8.1F A9.1J

2.1 Expressions and formulae *continued*

Subject content		Guidance	Curriculum reference
H	Simplify algebraic expressions using the index laws	$t \times t \times t = t^3$ $\frac{y^4 \times y^6}{y^5}$ $(2x^3)^3 = 8x^9$	A8.1A A8.1B
I	Write expressions and formulae involving one or more variables	A hotel charges $\$60$ per night to rent a room, plus a surcharge of $\$10$ per person per stay. Write a formula to show the relationship between the cost, C, the number of nights, n, and the number of people, p	A7.1H A9.1B
J	Substitute values into a formula and find the value of a variable that is not the subject	$F = ma$ If $F = 8.4$ and $a = 70$, find m	A7.1F A7.1G A8.1E A9.1C
K	Change the subject of a formula, involving the four operations and/or powers and roots	Make t the subject of the formula $Y = \frac{3t + 7}{4}$	A9.1D
L	Use index notation and index laws with positive, negative and zero powers	Write $\frac{64x^3}{16x^0}$ in its simplest form	N7.1J N8.1C N9.1C A9.1F
M	Expand the product of two linear expressions	$(x + 3)(x - 4)$	A9.1G
N	Factorise quadratic expressions of the form $x^2 + bx + c$	Factorise $x^2 - 7x - 30$	A9.1H
O	Factorise the difference of two squares	$x^2 - 36 = (x - 6)(x + 6)$	A9.1K

2.1 Expressions and formulae *continued*

Subject content		Guidance	Curriculum reference
P	Solve problems in context, involving formulae and expressions	The perimeter, P, of a rectangle can be found using the formula: $P = 2l + 2w$ Where l = length of the rectangle and w = width of the rectangle The perimeter of a rectangle is $156\,\text{cm}$ The length of the rectangle is twice as long as its width. What is the length of the rectangle?	A7.1I A8.1F A9.1J